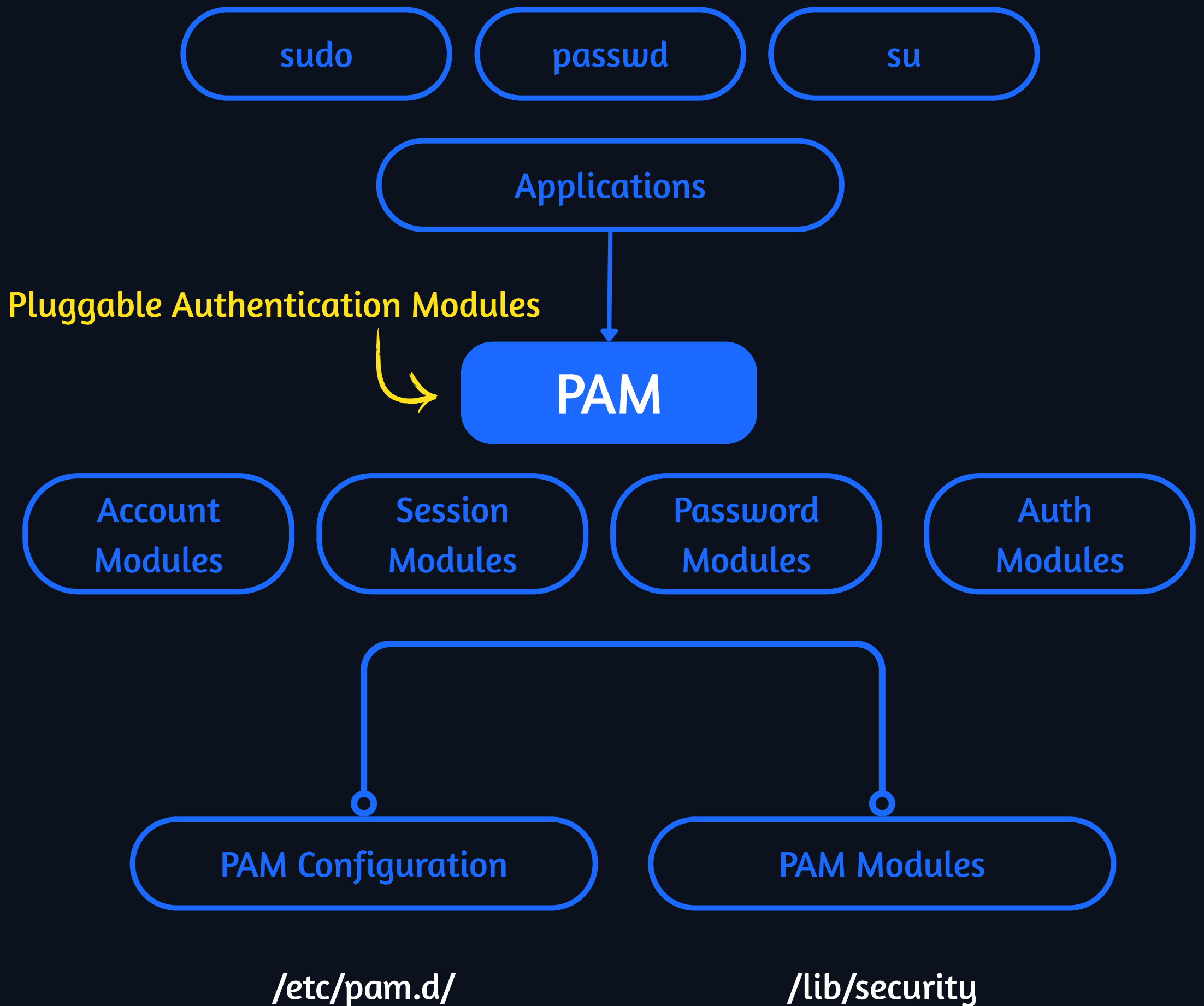


Mastering Linux Password Security: **PAM & pwquality.conf**

Want to enforce strong passwords in Linux?
Here's how `/etc/pam.d/common-password` and
`/etc/security/pwquality.conf` work together!

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What is PAM?



What is PAM?

PAM (Pluggable Authentication Modules) is a framework used in Linux to manage authentication tasks like logging in, changing passwords, or accessing services. Instead of hardcoding rules, PAM uses flexible, stackable modules defined in config files (like those in `/etc/pam.d/`).

When a user tries to authenticate, PAM checks a sequence of rules—each module handles a specific task like password checking or account restrictions—before granting or denying access.

How to enforce strong passwords using `/etc/pam.d/common-password`

To make sure users create secure passwords, Linux lets you define rules in the PAM stack. One of the key files is `/etc/pam.d/common-password`, which you can edit to control password strength. Here's how the process works:

- **Locate the file**

Open `/etc/pam.d/common-password` with a text editor like `nano` or `vim`:

```
sudo nano /etc/pam.d/common-password
```

- **Add or edit the line with `pam_pwquality.so`**

This module checks how strong a password is.

- **Edit the line to enforce rules**

For example:

```
password requisite pam_pwquality.so retry=3
```

How to enforce strong passwords using `/etc/pam.d/common-password`

- Fine-tune the password rules in
`/etc/security/pwquality.conf`

Instead of cramming all options into the PAM config, you can define rules cleanly in this file. For example:

`minlen = 12`

`ucredit = -1`

`lcredit = -1`

`dcredit = -1`

`ocredit = -1`

`enforce_for_root = 1`

- Save and exit

Press Ctrl+O, then Enter, and Ctrl+X to save changes and exit nano. Try changing a password with `passwd` to make sure the rules are applied.

What it does:

- **retry=3** – Allows the user 3 tries to enter a valid password.
- **minlen=12** – Requires passwords to be at least 12 characters long.
- **ucredit=-1** – Requires at least 1 uppercase letter.
- **lcredit=-1** – Requires at least 1 lowercase letter.
- **dcredit=-1** – Requires at least 1 digit.
- **ocredit=-1** – Requires at least 1 special character.
- **enforce_for_root** – Applies the same rules even to the root user.

Note:

Make sure the **libpam-pwquality** package is installed on your system. It provides the `pam_pwquality.so` module. You can install it with:

```
sudo apt install libpam-pwquality
```

Without it, password quality checks won't work.